

WM — What's in a Medicine?

OCR B (Salters) H433 storyline **WM**. From willow bark to aspirin: the chemistry of the -OH group in alcohols and phenols, carboxylic acids and esters, IR and mass spectrometry, preparative techniques, chromatography and green chemistry.

1. Alcohols

Classification

Primary alcohol (1°): The -OH group is attached to a carbon bonded to one other carbon (or none, as in methanol).

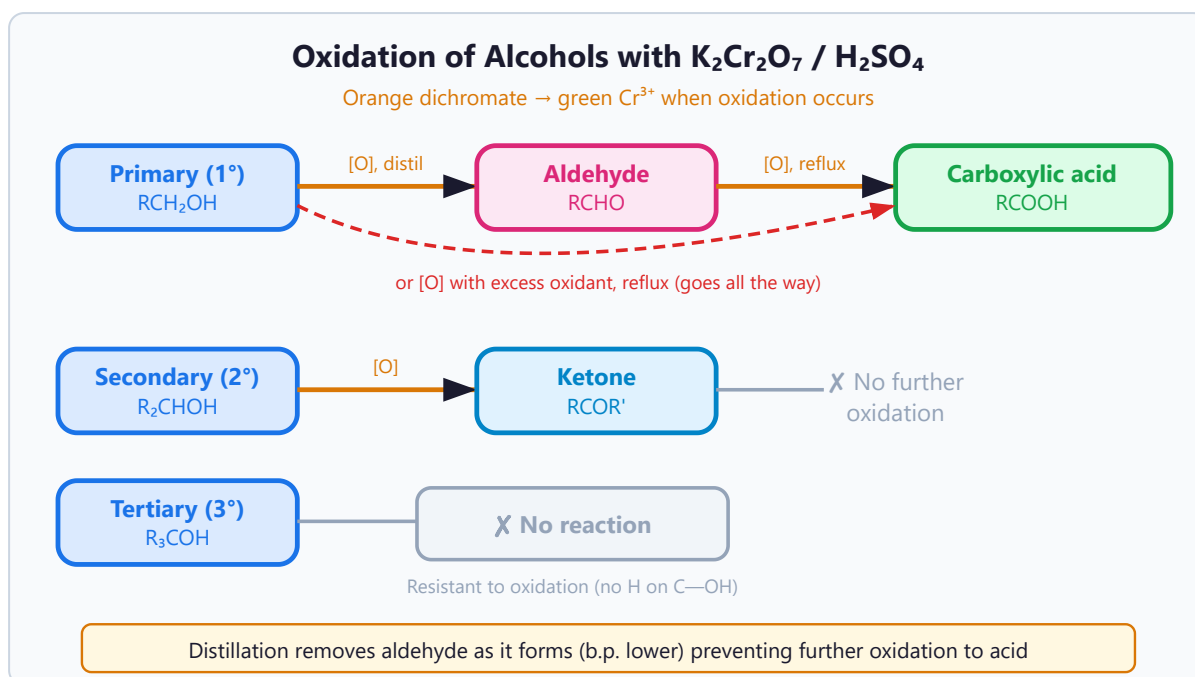
Secondary alcohol (2°): The -OH group is attached to a carbon bonded to two other carbons.

Tertiary alcohol (3°): The -OH group is attached to a carbon bonded to three other carbons.

Oxidation of Alcohols

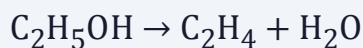
The oxidising agent is acidified potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}_2\text{SO}_4$). The dichromate changes from orange to green ($\text{Cr}_2\text{O}_7^{2-}$ to Cr^{3+}) when oxidation occurs. The oxidising agent is represented as [O] in equations.

Alcohol type	Product with distillation	Product with reflux
Primary (1°)	Aldehyde (distil immediately to prevent further oxidation)	Carboxylic acid
Secondary (2°)	Ketone (cannot be further oxidised)	
Tertiary (3°)	No reaction (resistant to oxidation under these conditions)	



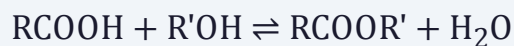
Dehydration

Alcohols can be dehydrated to form alkenes by heating with a strong acid catalyst (concentrated H_2SO_4 or H_3PO_4) or by passing alcohol vapour over heated aluminium oxide (Al_2O_3). Water is eliminated:



Esterification

Alcohols react with carboxylic acids in the presence of a concentrated sulfuric acid catalyst to form esters and water (a condensation reaction):



This is a reversible reaction. Esters have characteristic fruity smells and are used as flavourings, solvents, and in making plasticisers.

2. Carboxylic Acids and Derivatives

Carboxylic Acids

Carboxylic acids contain the -COOH functional group. They are weak acids that partially dissociate in water: $\text{RCOOH} \rightleftharpoons \text{RCOO}^- + \text{H}^+$. They react with bases, carbonates, and reactive metals to form salts (carboxylates).

Esters

Esters (RCOOR') are formed by the condensation reaction between a carboxylic acid and an alcohol (Fischer esterification) in the presence of an acid catalyst (concentrated H_2SO_4). This is a reversible reaction.

Hydrolysis of esters:

- **Acid hydrolysis:** Refluxing with dilute acid (e.g. dilute H_2SO_4) breaks the ester back into the carboxylic acid and alcohol. This is reversible.
- **Base hydrolysis (saponification):** Refluxing with dilute NaOH produces the sodium salt of the carboxylic acid (carboxylate) and the alcohol. This is irreversible because the carboxylate ion is very stable.

Acyl Chlorides

Acyl chloride (RCOCl): A reactive derivative of carboxylic acids in which the OH of the COOH group is replaced by Cl . They are much more reactive than carboxylic acids because the C-Cl bond is weaker and the Cl is a good leaving group.

Acyl chlorides react vigorously at room temperature with:

Reagent	Product	Other product
Water (H_2O)	Carboxylic acid	HCl
Alcohol (ROH)	Ester	HCl
Ammonia (NH_3)	Amide (RCONH_2)	HCl
Amine ($\text{R}'\text{NH}_2$)	N-substituted amide (RCONHR')	HCl

These are all nucleophilic addition-elimination reactions. The nucleophile (H_2O , ROH , NH_3 , or $\text{R}'\text{NH}_2$) attacks the δ^+ carbonyl carbon, and the Cl^- leaves.

3. Organic Synthesis and Retrosynthesis

Organic synthesis: Planning a series of reactions to convert a starting material into a desired target molecule.

Retrosynthesis: Working backwards from the target molecule to identify simpler starting materials and the reactions needed. The target is disconnected at key bonds to identify precursor molecules.

Key synthetic routes to know (building a "reaction map"):

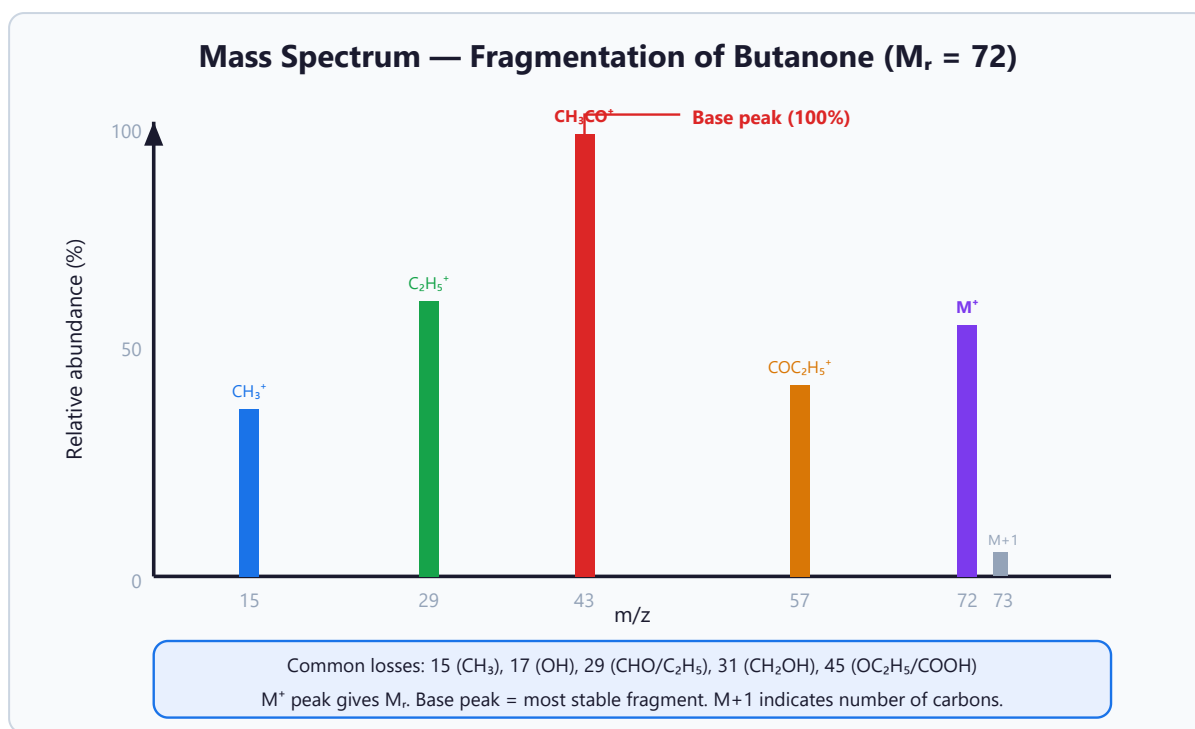
- Alkene $\rightarrow$ alcohol (hydration with H_3PO_4 catalyst, or addition of H_2O via H_2SO_4)
- Alcohol $\rightarrow$ halogenoalkane (HBr , or $\text{NaBr} + \text{H}_2\text{SO}_4$, or $\text{PCl}_3/\text{PCl}_5/\text{SOCl}_2$)
- Halogenoalkane $\rightarrow$ amine (excess NH_3 , sealed tube)
- Halogenoalkane $\rightarrow$ nitrile (KCN in ethanol/water) $\rightarrow$ carboxylic acid (acid hydrolysis) or amine (LiAlH_4 reduction)
- Alcohol $\rightarrow$ aldehyde (oxidation, distil) $\rightarrow$ carboxylic acid (further oxidation, reflux)
- Carboxylic acid + alcohol $\rightarrow$ ester (acid catalyst)
- Acyl chloride + amine $\rightarrow$ amide
- Benzene $\rightarrow$ nitrobenzene (nitration) $\rightarrow$ phenylamine (reduction with Sn/HCl)

4. Analytical Techniques

Mass Spectrometry

Mass spectrometry identifies compounds by their molecular mass and fragmentation pattern. The molecular ion peak (M^+) gives the relative molecular mass. Fragmentation peaks correspond to pieces of the molecule after bonds break. Common fragment losses: 15 (CH_3), 17 (OH), 29 (CHO or C_2H_5), 31 (CH_2OH), 45 (OC_2H_5 or COOH).

The $\text{M}+1$ peak (one mass unit above M^+) is caused by the presence of ^{13}C . The relative height of the $\text{M}+1$ peak indicates the number of carbon atoms in the molecule.

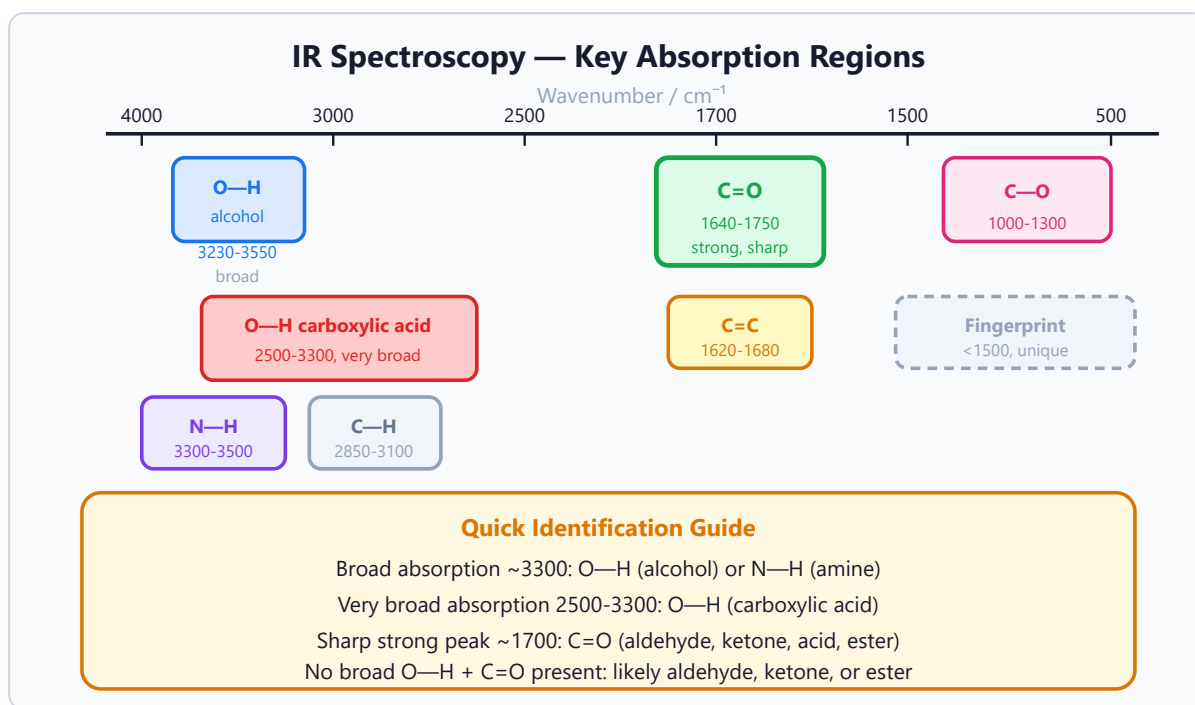


Infrared (IR) Spectroscopy

IR spectroscopy: Identifies functional groups by the absorption of infrared radiation, which causes covalent bonds to vibrate (stretch and bend) at characteristic frequencies. An IR spectrum plots transmittance (%) against wavenumber (cm^{-1}).

Bond	Wavenumber (cm^{-1})	Appearance
O–H (alcohol)	3230–3550	Broad
O–H (carboxylic acid)	2500–3300	Very broad
N–H (amine)	3300–3500	Medium, may show 2 peaks
C–H	2850–3100	Medium to strong
C=O	1640–1750	Strong, sharp
C=C	1620–1680	Medium
C–O	1000–1300	Strong

The fingerprint region (below about 1500 cm^{-1}) is unique to each compound and can be compared against a database for identification. IR spectroscopy can also be used to monitor the progress of a reaction by observing the disappearance or appearance of characteristic absorptions.



NMR Spectroscopy

Nuclear Magnetic Resonance (NMR) spectroscopy: Detects the chemical environments of specific nuclei (usually ^1H or ^{13}C) in a magnetic field. Nuclei in different chemical environments absorb at different frequencies, giving peaks at different chemical shifts (δ , measured in ppm relative to TMS).

Tetramethylsilane (TMS, $\text{Si}(\text{CH}_3)_4$) is the reference standard ($\delta = 0\text{ ppm}$) because it is chemically inert, gives a single sharp peak (all 12 H atoms are equivalent), and its peak is well away from most other signals.

^{13}C NMR

^{13}C NMR shows one peak for each unique carbon environment in the molecule. The number of peaks tells you how many different carbon environments there are. Symmetry can reduce the number of peaks (equivalent carbons give only one peak). Chemical shift values indicate the type of carbon environment:

Carbon environment	Approximate δ / ppm
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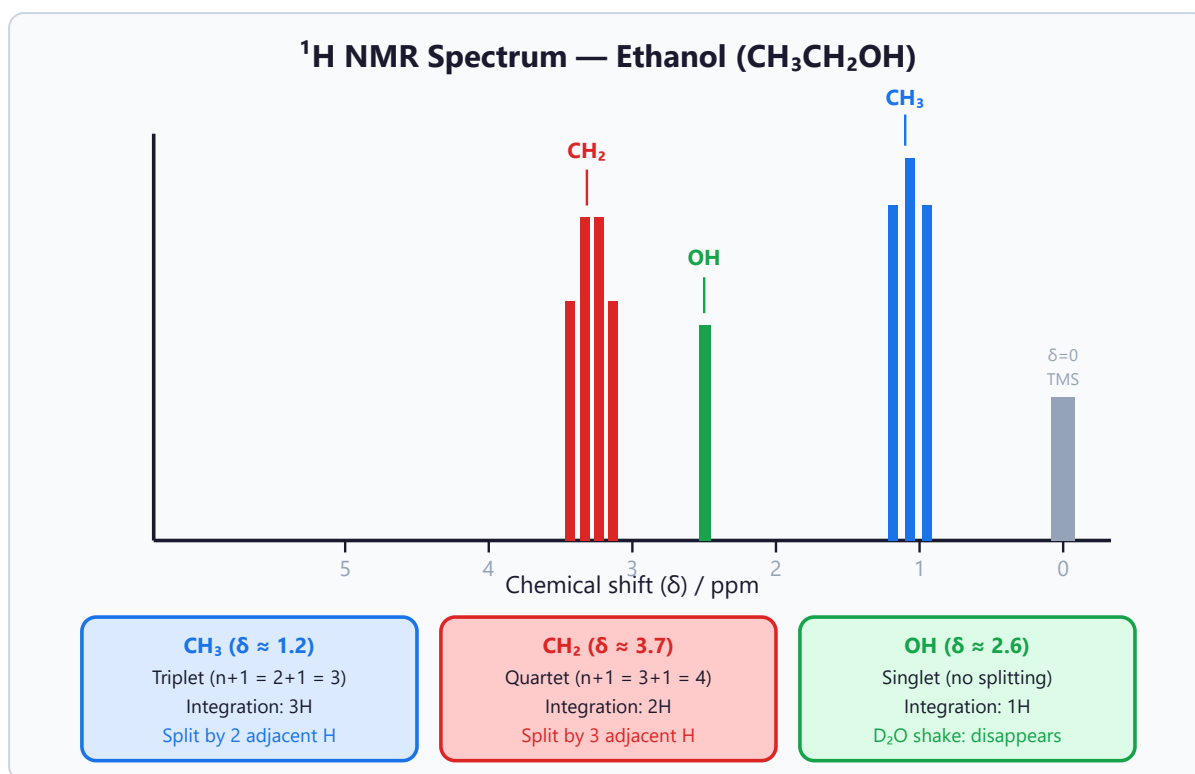
R-CH ₃ , R-CH ₂ -R (alkyl)	5–40
C-Cl, C-Br	25–65
C-O (alcohols, esters)	50–80
C=C (aromatic)	110–160
C=O (carbonyl)	170–220

¹H NMR

¹H NMR is richer in information than ¹³C NMR because it provides three pieces of data for each signal:

1. **Chemical shift (δ):** Indicates the chemical environment of the hydrogen atoms.
2. **Integration (relative peak area):** Proportional to the number of hydrogen atoms in that environment.
3. **Splitting pattern (multiplicity):** Determined by the number of hydrogen atoms on adjacent carbon atoms (the n+1 rule).

Proton environment	Approximate δ / ppm
R-CH ₃	0.7–1.2
R-CH ₂ -R	1.2–1.4
R-CH ₂ -X (halogen or O)	3.1–4.2
R-O-H (alcohol)	1.0–5.5 (variable)
Ar-H (aromatic)	6.5–8.0
R-CHO (aldehyde)	9.4–10.0
R-COOH (carboxylic acid)	10.0–12.0



Splitting (Spin-Spin Coupling)

The **$n+1$ rule**: A proton signal is split into $(n+1)$ peaks, where n is the number of equivalent protons on adjacent carbon atoms.

Adjacent equivalent H atoms (n)	Splitting pattern	Name
0	1 peak	Singlet (s)
1	2 peaks	Doublet (d)
2	3 peaks	Triplet (t)
3	4 peaks	Quartet (q)

The relative intensities follow Pascal's triangle: singlet 1, doublet 1:1, triplet 1:2:1, quartet 1:3:3:1. O–H and N–H protons usually appear as singlets (no splitting) because rapid exchange with the solvent averages out coupling. A D₂O shake (adding deuterium oxide) causes O–H and N–H peaks to disappear, confirming their identity.

Exam Tip: When interpreting ^1H NMR spectra: (1) count the number of signals to determine how many different H environments exist; (2) use integration ratios to find the relative number of H atoms; (3) use chemical shifts to identify the environments; (4) use splitting patterns to determine connectivity. Work systematically through all the data before proposing a structure.

5. Chromatography

Thin-Layer Chromatography (TLC)

TLC uses a thin layer of silica gel or alumina on a plate (the stationary phase). The sample is spotted near the bottom, and a solvent (mobile phase) rises up the plate by capillary action. Components separate based on their relative affinities for the stationary and mobile phases.

$$R_f = \text{distance moved by substance} / \text{distance moved by solvent front}$$

R_f values are characteristic of a substance under specific conditions and can be compared with standards for identification. TLC is quick and useful for monitoring reaction progress and checking purity.

Gas Chromatography (GC)

GC separates volatile compounds. The sample is vaporised and carried through a long, coiled column by an inert carrier gas (the mobile phase). The column is packed with a high-boiling-point liquid on an inert solid support (the stationary phase). Components separate based on their boiling points and interactions with the stationary phase. A detector records the signal as each component elutes, producing a chromatogram where peak retention time identifies the component and peak area is proportional to amount.

High-Performance Liquid Chromatography (HPLC)

HPLC is used for non-volatile or thermally unstable compounds that cannot be analysed by GC. A liquid mobile phase is pumped at high pressure through a column packed with fine particles (stationary phase). The technique gives excellent separation and is widely used in pharmaceutical analysis. A UV detector is commonly used. Like GC, retention time identifies components and peak area gives quantitative information.

GC-MS and HPLC-MS combine chromatographic separation with mass spectrometry for both separation and identification in a single analysis, making them extremely powerful analytical tools.

Exam Tip: You may be asked to combine data from mass spectrometry, IR, and NMR to identify an unknown compound. Approach this systematically: (1) use the mass spectrum to find the molecular mass and key fragments; (2) calculate the degree of unsaturation ($\text{IHD} = (2C + 2 + N - H - X) / 2$; IHD of 4 strongly suggests a benzene ring); (3) use IR to identify functional groups; (4) use ^{13}C NMR to count carbon environments; (5) use ^1H NMR to determine hydrogen environments, numbers, and connectivity. Propose a structure consistent with all the data.

Worked Example — Structure Determination:

Unknown compound: $M^+ = 72$. Molecular formula $\text{C}_4\text{H}_8\text{O}$. $\text{IHD} = (8 + 2 - 8) / 2 = 1$ (one $\text{C}=\text{O}$ or one ring).

IR: strong sharp absorption at 1720 cm^{-1} ($\text{C}=\text{O}$), no broad $\text{O}-\text{H}$ → ketone or aldehyde, not acid or alcohol.

^1H NMR: three peaks. δ 1.0 (triplet, 3H), δ 2.1 (singlet, 3H) and δ 2.4 (quartet, 2H). No peak at δ 9–10, so not an aldehyde.

The 3H singlet at δ 2.1 is a CH_3 next to $\text{C}=\text{O}$ with no adjacent H atoms; the 2H quartet and 3H triplet show a CH_2CH_3 group, with the CH_2 next to $\text{C}=\text{O}$.

Structure: butanone ($\text{CH}_3\text{COCH}_2\text{CH}_3$). Always cross-check all data against your proposed structure.